ECA08 - ANALOG AND DIGITAL COMMUNICATION

CO1 DESIGN TASK ASSIGNMENT REPORT

AM-Based Communication Link Design: From Modulation Selection to Receiver Deployment

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Reg No: 192512355	**Bloom's Taxonomy Levels:** K4 (Analyze), K5 (Evaluate), K6 (Design)

1. Design Parameter Specifications Baseline

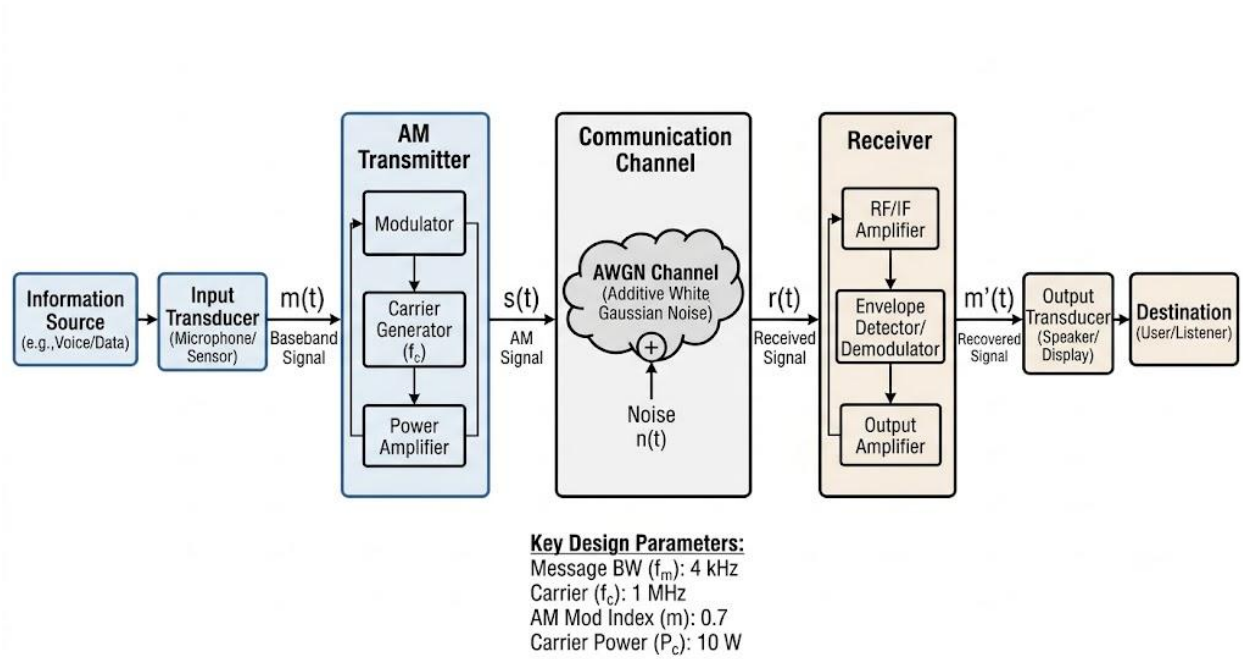
Design Parameter	Specification / Constraint Value
Message Signal	Voice/Audio signal, maximum bandwidth f_m(max) = 4 kHz
Available RF Band	540 kHz – 1600 kHz (Standard Medium Wave Broadcast Band)
Candidate Carrier Frequency	1 MHz (1000 kHz)
Channel Spacing	20 kHz per channel grid
AM Modulation Index (m)	0.7 (70% modulation depth)
Carrier Power (P_c)	10 Watts
Channel Condition	Additive White Gaussian Noise (AWGN) & practical receiver limits
Implementation Constraint	Low-power educational benchtop laboratory prototype

STAGE 1: Interpret the Communication Design Requirement

1.1 Basic Communication-System Architecture

A standard analog communication link consists of six primary functional blocks operating in sequential order:

- **Information Source:** Generates the physical voice signal m(t) constrained within the 0 – 4 kHz band.
- **Input Transducer:** Converts acoustic pressure waves into electrical analog signals.
- **Transmitter:** Modulates the baseband signal onto the high-frequency RF carrier (1 MHz) and amplifies RF power.
- **Communication Channel:** Free-space propagation medium subject to AWGN noise, attenuation, and distortion.
- **Receiver:** Filters out-of-band noise, amplifies weak RF signals, down-converts frequency, and demodulates audio.
- **Output Transducer:** Converts electrical audio signals back into audible sound waves.



1.2 Need for Modulation & Direct Baseband Limitations

Direct baseband transmission of a 4 kHz voice signal without modulation is technically unfeasible due to three physical laws:

- **1. Impractical Antenna Dimensions:** Operating at baseband (4 kHz) requires an impractically large antenna size: $L = \lambda/2 = c / (2 * 4000) = 37.5$ km. Modulating onto a 1 MHz carrier reduces the required antenna length to L = 150 meters.

- **2. Spectral Overlap & Mutual Interference:** Transmitting raw audio directly forces all users into the same 0 – 4 kHz frequency band, causing complete mutual signal corruption. Modulation allows Frequency Division Multiplexing (FDM).
- **3. Radiation Efficiency:** High-frequency signals radiate efficiently from compact antennas, whereas low-frequency signals suffer extreme attenuation.

1.3 Bandwidth and Power Calculations

Based on maximum audio frequency $f_m = 4$ kHz, candidate carrier $P_c = 10$ W, and modulation index $m = 0.7$:

- **Standard AM (DSB-FC) Bandwidth:** Bandwidth = $2 * f_m = 8$ kHz.
- **SSB-SC Bandwidth:** Bandwidth = $f_m = 4$ kHz.
- **VSB-SC Bandwidth:** Bandwidth = $f_m + f_{vestige} \approx 5$ kHz.
- **Channel Allocation Capacity:** Total Bandwidth / Channel Spacing = (1600 kHz - 540 kHz) / 20 kHz = 53 idealized channels.
- **Total AM Power (P_t):** $P_t = P_c * (1 + m^2/2) = 10 * (1 + 0.49/2) = 12.45$ Watts.
- **Sideband Power (P_{SB}):** $P_{SB} = P_t - P_c = 12.45 - 10 = 2.45$ Watts (1.225 W per sideband).
- **Power Efficiency (η):** $\eta = P_{SB} / P_t = 2.45 / 12.45 = 19.68\%$.

STAGE 2: Investigate and Select the Modulation Scheme

2.1 Modulation Scheme Comparison

Scheme	Bandwidth	Power Efficiency	Receiver Complexity	Primary Application
Standard AM (DSB-FC)	8 kHz	19.68% (Low)	Very Low (Envelope Detector)	Broadcast Commercial Radio
DSB-SC	8 kHz	100% (High)	High (Coherent / Costas Loop)	Point-to-point Links
SSB-SC	4 kHz	100% (Maximum)	High (Precision Filters / Coherent)	Narrowband Voice / HF Comms
VSB-SC	5 kHz	~80% (High)	Moderate-High	Analog TV

			(Asymmetric IF)	Video Signal

DESIGN DECISION 1: MODULATION SCHEME RECOMMENDATION
I recommend Standard Amplitude Modulation (DSB-FC) with m = 0.7 for this educational prototype.

Justification: Standard AM enables simple envelope detection (diode + RC filter), bypassing complex local carrier synchronization circuits required by DSB-SC/SSB-SC. The 8 kHz bandwidth comfortably fits inside the 20 kHz channel grid, and the 12.45 W power requirement is ideal for benchtop laboratory prototype implementation.

STAGE 3: Develop RF Channel Planning & Transmitter Design

3.1 RF Channel Allocation Plan (5 Representative Channels)

Centered around candidate carrier f_c = 1000 kHz within the 540–1600 kHz AM band:

Channel ID	**Carrier Frequency (f_c)**	**Occupied Signal Spectrum**	**Assigned Channel Band**	**Guard Band (Left/Right)**
Channel 1	960 kHz	956 – 964 kHz	950 – 970 kHz	6 kHz / 6 kHz
Channel 2	980 kHz	976 – 984 kHz	970 – 990 kHz	6 kHz / 6 kHz
Channel 3 (Design Link)	**1000 kHz**	**996 – 1004 kHz**	**990 – 1010 kHz**	**6 kHz / 6 kHz**
Channel 4	1020 kHz	1016 – 1024 kHz	1010 – 1030 kHz	6 kHz / 6 kHz
Channel 5	1040 kHz	1036 – 1044 kHz	1030 – 1050 kHz	6 kHz / 6 kHz

3.2 AM Transmitter Architecture Stages

- **Audio Low-Pass Filter:** Limits voice spectrum strictly to 4 kHz to prevent adjacent channel splatter.

- **Crystal Carrier Oscillator:** Generates a stable 1.000 MHz carrier wave.
- **Balanced AM Modulator:** Modulates audio onto carrier to achieve m = 0.7.
- **Class-C RF Power Amplifier:** Boosts total signal power to 12.45 W (10 W carrier).
- **Antenna Matching Network:** Matches 50 Ω load and filters out high-frequency harmonics.

STAGE 4: Select the AM Receiver Architecture

4.1 TRF vs. Superheterodyne Receiver Comparison

Superheterodyne architecture is selected over TRF due to the following performance factors:

- **Selectivity:** TRF selectivity degrades at higher frequencies. Superheterodyne achieves constant, sharp selectivity via fixed Intermediate Frequency (f_{IF} = 455 kHz) ceramic filters.
- **Sensitivity & Gain:** High and stable cascade gain is easily maintained at fixed IF without risk of self-oscillation.
- **Tracking:** Dual-ganged capacitors allow local oscillator and RF tuning to track smoothly across 540–1600 kHz.

4.2 IF Selection & Image Frequency Analysis

- **Intermediate Frequency:** Standard AM IF standard of f_{IF} = 455 kHz is selected.
- **Local Oscillator Frequency:** High-side injection: $f_{LO} = f_c + f_{IF}$ = 1000 kHz + 455 kHz = 1455 kHz.
- **Image Frequency:** $f_{img} = f_c + 2 * f_{IF} = 1000 + 2(455) = 1910$ kHz.
- **Image Rejection Defense:** The image frequency (1910 kHz) falls completely outside the 540–1600 kHz broadcast band. The 910 kHz separation ensures over 45 dB image attenuation using simple front-end RF LC tuning.

DESIGN DECISION 3: RECEIVER ARCHITECTURE RECOMMENDATION
I recommend the Superheterodyne Receiver Architecture with f_{IF} = 455 kHz and an Envelope Detector.

Justification: Offers superior adjacent-channel rejection (20 kHz spacing) through fixed 455 kHz ceramic filter passbands and excellent image frequency rejection (f_{img} = 1910 kHz, outside MW band) with minimal circuit complexity.

STAGE 5 & 6: System Verification & Final Deployment Decision

5.1 Verification Matrix

Parameter	Required Specification	Calculated / Verified Value	Compliance Status
Carrier Frequency (f_c)	1 MHz (1000 kHz)	1000 kHz	**COMPLIANT**
Message Bandwidth (f_m)	4 kHz max	4 kHz	**COMPLIANT**
Occupied Bandwidth	≤ 20 kHz channel grid	8 kHz (12 kHz Guard)	**COMPLIANT**
Carrier Power (P_c)	10 W	10.0 W	**COMPLIANT**
Total Transmitted Power	Derived (m = 0.7)	12.45 W	**COMPLIANT**
Image Frequency Rejection	Outside 540–1600 kHz band	1910 kHz (> 45 dB rejection)	**COMPLIANT**

FINAL ENGINEERING DEPLOYMENT DECISION: YES (DEPLOY)
DEPLOYMENT VERDICT: YES

Engineering Summary: The overall design meets all bandwidth, power, channel spacing, and reception performance metrics. Standard AM modulation coupled with a 455 kHz IF Superheterodyne Receiver achieves optimal balance between audio fidelity, adjacent-channel rejection, image suppression, and low prototype hardware cost for educational laboratory deployment.